

**THERMAL MODELING OF $\text{LaNi}_{4.91}\text{Sn}_{0.15}$ BASED SOLID
STATE HYDROGEN STORAGE DEVICE WITH COPPER
FINNED EMBEDDED COOLING TUBES (ECT)**

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF

**BACHELOR OF TECHNOLOGY
IN
MECHANICAL ENGINEERING**

BY

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UNDER THE GUIDANCE OF
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CERTIFICATE

This is to certify that this thesis entitled, “**Thermal Modelling of $\text{LmNi}_{4.91}\text{Sn}_{0.15}$ Based Solid State Hydrogen Storage Device with Copper Finned Embedded Cooling Tubes (ECT)**” submitted by **U Aditya Mohan (117ME0504)**, in partial fulfilments for the requirements for the degree of **Bachelor of Technology in Mechanical Engineering** at **National Institute of Technology, Rourkela** is an authentic work carried out by him under my supervision and guidance. To the best of my knowledge, the matter embodied in the thesis has not been submitted to any other University/Institutes for the award of any Degree or Diploma.

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ABSTRACT

Hydrogen is considered as one of the most effective and eco-friendly energy carriers, whereas the storage of Hydrogen has been a challenge because of its low-density, hence it requires high-grade of energy to store it in form of pressurized gas or cryogenic liquid. There comes Metal Hydride based storage device which can store Hydrogen safely and efficiently but it takes a long charging and discharging time which limits its use to certain fields. To predict the minimum time for charging or discharging of Hydrogen in Metal Hydride based Hydrogen storage device, a 2-D mathematical model is developed with 60 ECT to give a basic understanding of how cooling and absorption take place in the alloy using COMSOL Multiphysics 5.6. This study is then extended to a 3-D mathematical model with 36 ECT. The performance of Hydrogen storage device during the Hydriding process is presented for (4-10) number of fins, the thickness of fin (1mm-3mm), and convective heat transfer coefficient of the shell (0-2000 W/m²K) at a supply Hydrogen pressure of 30bar and cooling fluid temperature at the inlet as 20°C. It is observed that as the number of fins increases the absorption time decrease keeping the fin thickness same and also if fin thickness increases the absorption rate increases. From different graphs plotted, it is observed that 10 fins each of thickness 1mm can enhance the performance of the Hydrogen storage device with minimal increase in weight of the device. hence the optimal design is finalized.

Keywords: Metal Hydride based Hydrogen Storage Device, Heat transfer Fluid, Absorption.

LIST OF CONTENT

Contents	Page no.
<i>CERTIFICATE</i>	
<i>ACKNOWLEDGEMENT</i>	
<i>ABSTRACT</i>	
Chapter 1 INTRODUCTION	
1.1 Motivation	2
1.2 Objectives	2
Chapter 2 LITERATURE SURVEY	4
Chapter 3 METHODOLOGY	
3.1 Mathematical model	6
3.2 Formulations	7
Chapter 4 RESULTS AND CONCLUSION	
4.1 Result and Discussion	10
4.2 Conclusion	15
4.3 Future scope	15
<i>REFERENCES</i>	16

Chapter 1

Introduction

1. INTRODUCTION

1.1 MOTIVATION

The world currently faces issues related to Global warming, fossil fuel depletion, etc. The consequences severely affect the future, therefore researchers are taking enormous efforts to use renewable and cleaner fuels in the industries and for on-board applications. Hydrogen has been considered as one of the most effective and eco-friendly energy carrier. As hydrogen has is a very low-density gas, it requires high-graded energy to store it in the form of pressurized gas or cryogenic liquid hydrogen. Metal Hydride (MH) based hydrogen storage devices is one of the safest and reliable method of storing hydrogen efficiently as it requires only low-grade energy [1], hence more attention is being paid to the development of MH based storage device leading to widening its application in the field of heat transformer, fuel cell, hydrogen compressor, etc. But the poor thermal conductivity of MH bed and its high heat of adsorption during charging has led to large charging times. Therefore, heat exchanger designs are needed to be modified to increase the heat transfer rate between MH bed and heat transfer fluid (HTF).

1.2 OBJECTIVES

In this experiment conducted, a 3-D model of $\text{LaNi}_{4.91}\text{Sn}_{0.15}$ alloy-based hydrogen storage device with 36 copper finned ECT is designed which predicts the characteristics of absorption in it using COMSOL Multiphysics 5.6. And improving the model design to decrease the time taken for the hydriding by changing the number of fins used, changing the thickness of fin used, and convective heat transfer coefficient of the shell.

Chapter 2

Literature Survey

2. LITERATURE REVIEW

With increasing need for the use of Hydrogen, researchers are concentrating their research towards the development of Metal Hydride Based Hydrogen Storage Device (MHHSD) as its most reliable, safe and efficient way of storing Hydrogen at a temperature close to ambient temperature so that it can easily be used in vehicles or in large-scale industries [2].

Anbarasu et al. [3,4] developed a 2-D mathematical model varying the number of Embedded Cooling Tubes (ECT), supply pressure of Hydrogen, Temperature of HTF, and Thickness of Hydriding bed for evaluating the hydriding characteristics of MHHSD. The study was also extended to 3-D mathematical model for predicting the hydriding characteristics of $\text{LaNi}_{4.91}\text{Sn}_{0.15}$ with 60 ECT.

Ha et al. [5] for predicting the hydriding characteristics, developed a 2-D mathematical model of LaNi_5 based MHHSD. The research proved that the heat transfer rate increases for increasing thermal conductivity of the bed, decreasing thickness of bed. Also they reported significant increase in heat rate with addition of fins in the MH bed.

Muthukumar and Ramana [6] evaluated the desorption characteristics during Hydrogen Desorption process by developing a 2-D numerical model of $\text{MmNi}_{4.6}\text{Al}_{0.4}$ based MHHSD at different operating conditions. The study proved the importance of temperature variation of HTF along the axial direction. And also the governing equation was formulated for the reactor.

Muthukumar et al. [7] studied the effect of various operating conditions like different temperature of HTF, supply pressure of the Hydrogen, and overall heat transfer coefficients using a 2-D mathematical model in COMSOL Multiphysics. Industrial scale LaNi_5 MHHSD reactor with 48 ECT was used for the experiment and its performance was evaluated considering 120-150 kg of alloy.

Chapter 3

Methodology

3. METHODOLOGY

3.1. MATHEMATICAL MODEL

A 2-D model was first developed in COMSOL Multiphysics 5.6 representing the cross-section of the alloy in MH based Hydrogen storage device. The circular cross-section is shown in **Fig.1(a)**, the circular section represents the alloy surface, the circular part at the center represents the filter from which hydrogen enters the alloy bed and the small circular pores represent the cooling tube through which water at 20°C is flowing.

The study is then extended to a 3-D mathematical model as shown in **Fig.1(b)**. The 3-D model includes fins, outer shell, water tubes, and water (Heat Transfer Fluid).

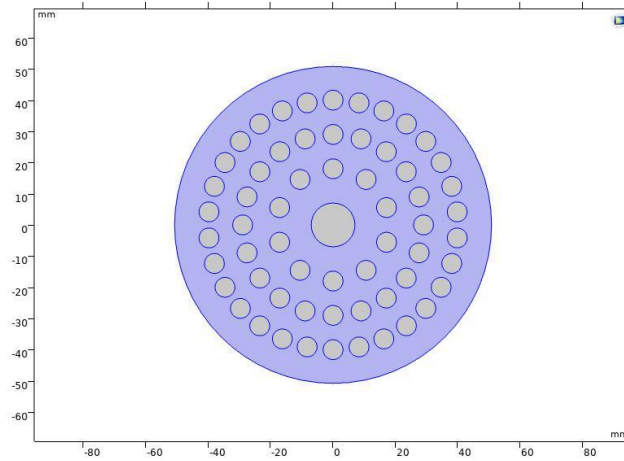


Fig.1(a). Cross-sectional view of MH based Hydrogen storage device with 60 ECT (2-D model)

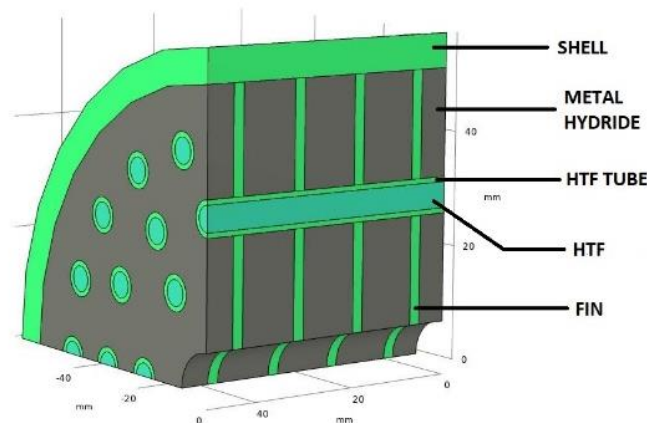


Fig.1(b). Sectional one fourth view of MH based Hydrogen storage device with 36 ECT (for 3-D model)

The design parameters are shown in **Table-1**. The solution procedure and the assumptions to simplify the governing equations are similar to the one presented in Muthukumar et al. [6]. The thermophysical properties of $\text{LaNi}_{4.91}\text{Sn}_{0.15}$ and hydrogen used to compute the present problem are listed in **Table-2**. The mesh of the computational domain considered is as free fine tetrahedral mesh and simulations are performed using COMSOL Multiphysics 5.6. Which can effectively solve the energy equation both convection and conduction mode heat transfer taking place inside the storage unit and the hydrogen mass (diffusion) and momentum (Darcy's law) transport equations in porous media module. User-defined functions are incorporated for calculating the specific parameters such as the rate of hydrogen absorption and hydride concentrations during hydriding processes.

3.2. FORMULATIONS

The heat generated in the absorption process or heat source is given by:

$$Q = \frac{\Delta H \zeta}{M_g}$$

Here ΔH denotes, Heat of formation for MH and M_g denotes the molar mass of hydrogen.

The amount of hydrogen absorbed which is directly associated with rate of reaction is written by Refs. [6,7]:

$$\zeta = C_a \exp\left(\frac{-E_a}{R_u T}\right) \ln\left(\frac{P_s}{P_{eq}}\right) (\rho_{ss} - \rho_t) (1 - \varepsilon)$$

Where C_a is the reaction constant for absorption, E_a is the activation energy for absorption and ε is the porosity of the MH bed.

The equilibrium pressure of hydride bed during hydriding and dehydriding processes are calculated using Van't Hoff equation and it's expressed as [8]:

$$\frac{P_{eq}}{P_o} = \exp\left[\frac{\Delta S}{R_u} - \frac{\Delta H}{R_u T} + (\varphi_s \pm \varphi_o) \times \tan\left(\pi\left(\frac{x}{x_f} - \frac{1}{2}\right)\right) \pm \frac{\beta}{2}\right]$$

Where “+” is for the absorption process and “-” is for the desorption process, ΔS is the entropy of formation, φ_s is the slope factor, X and X_f are concentration at the given time (t) and the final concentrations of H_2 AND β is hysteresis factor.

Shell Outer Radius	R_o	114.3/2[mm]
Shell Inner Radius	R_i	102.26/2[mm]
Filter Outer Radius	F_o	4/2[mm]
Tube Outer Radius	r_o	6.35/2[mm]
Tube Inner Radius	r_i	4.57/2[mm]
Length of Reactor	L	50[mm]
Number of fins	N_f	4
Distance of 1st layer tube from centre	r_1	18.08[mm]
Distance of 2nd layer tube from centre	r_2	29.15[mm]
Distance of 3rd layer tube from centre	r_3	40.225[mm]

Table-1 Design parameters of the MH based Hydrogen storage device used in the analysis

Initial bed Temp.	To	293[K]
Cooling Fluid Temp.	Tcool	293[K]
Ambient pressure	p_amb	1[bar]
Supplied Pressure	Ps	30[bar]
Gas Constant	Ru	8.314[J/mol/K]
Hydrogen Density	rho_g	0.0838[kg/m^3]
Molar Mass of Hydrogen	Mg	0.002[kg/mol]
Molar Mass of Alloy	M	0.432[kg/mol]
Weight fraction	Wt_sat	0.014
Porosity	e	0.5
Initial crystal Density of Bed	rho_i	8300[kg/m^3]
Reaction constant	Ca	59.187[1/s]
Activation Energy	Ea	21179.6[J/mol]
Enthalpy	H	30800[J/mol]
Entropy	S	108[J/mol/K]
Slope Factor Final	phi_s	0.038
Slope Factor Initial	phi_o	0
Hysteresis factor	Hys	0.137
Initial Concentration	Ci	0
Thermal conductivity of Hydrogen	K_g	0.127[W/(m*K)]
Velocity of water	v	0.1[m/s]

Table-2 Thermo-physical properties of LmNi_{4.95}Sn_{0.15}, hydrogen and constant used in the analysis

Chapter 4

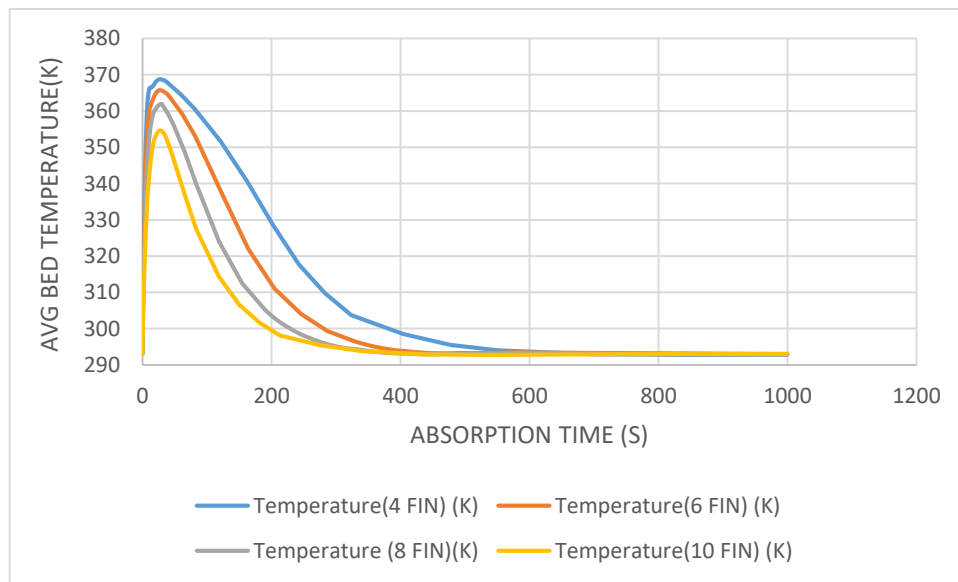
Results and Conclusion

4.1. RESULTS

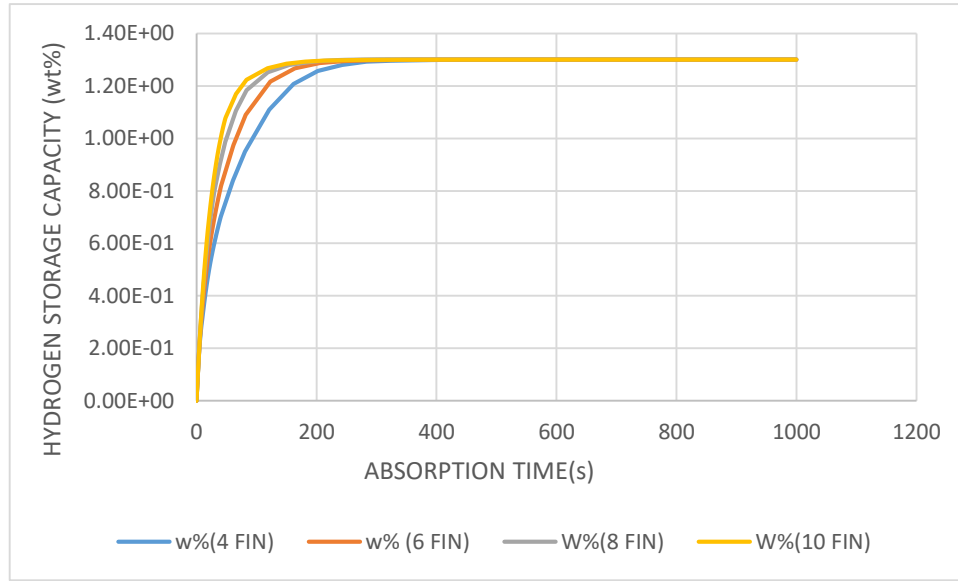
In this chapter, the effect on absorption rate in MH-based Hydrogen storage devices by changing the number of fins, thickness of fin, and Convective heat transfer coefficient of shell is discussed.

4.1.a. EFFECT OF NUMBER OF FINS

To obtain the lowest charging time, the number of fins was increased from 4 to 10 (4,6,8 and 10) keeping the thickness of the fin constant i.e. 2mm. It is observed that as the number of fin increases the maximum average bed temperature decreases as shown in **Fig.2(a)**, and as the temperature decreases the absorption rate will increase which is shown in **Fig.2(b)**. This happens because fins increase the net conductive heat transfer coefficient inside the bed and more amount of heat is transferred to HTF and the surrounding via the shell surface. But at the same time as the number of fin increases, the weight of the MH-based storage device will also increase which is undesirable. Therefore, the selection of optimum number of fins is necessary. It can be observed that the time taken in absorption by the device with 8 fins is very close to the device with 10 fins. Therefore, after 1st experiment, the storage device with 8 fins each of 2mm was considered



(a)

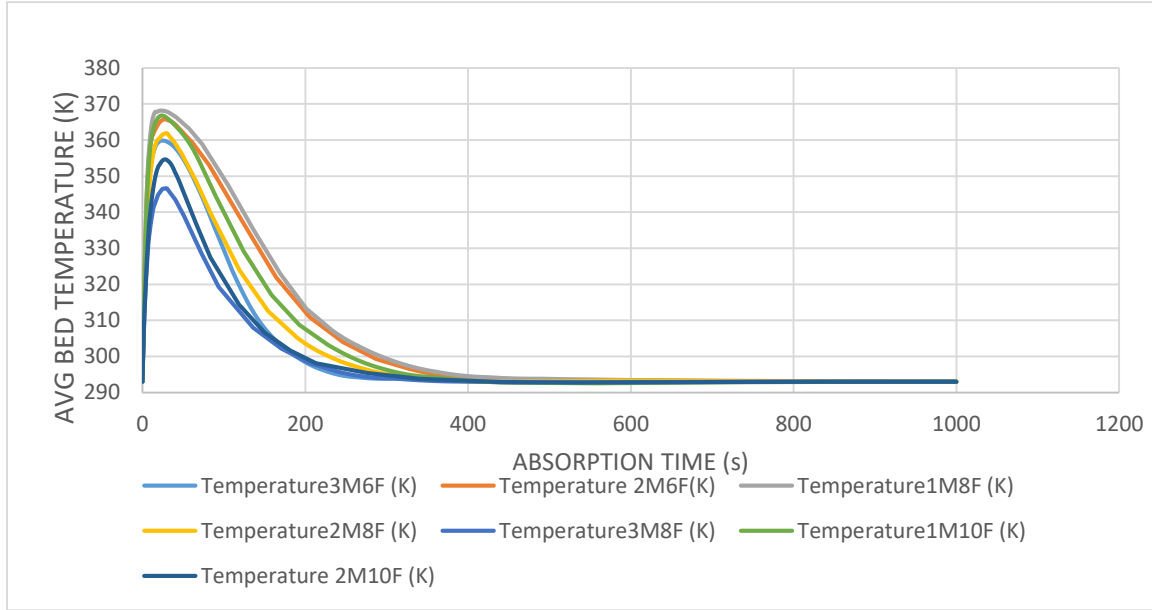


(b)

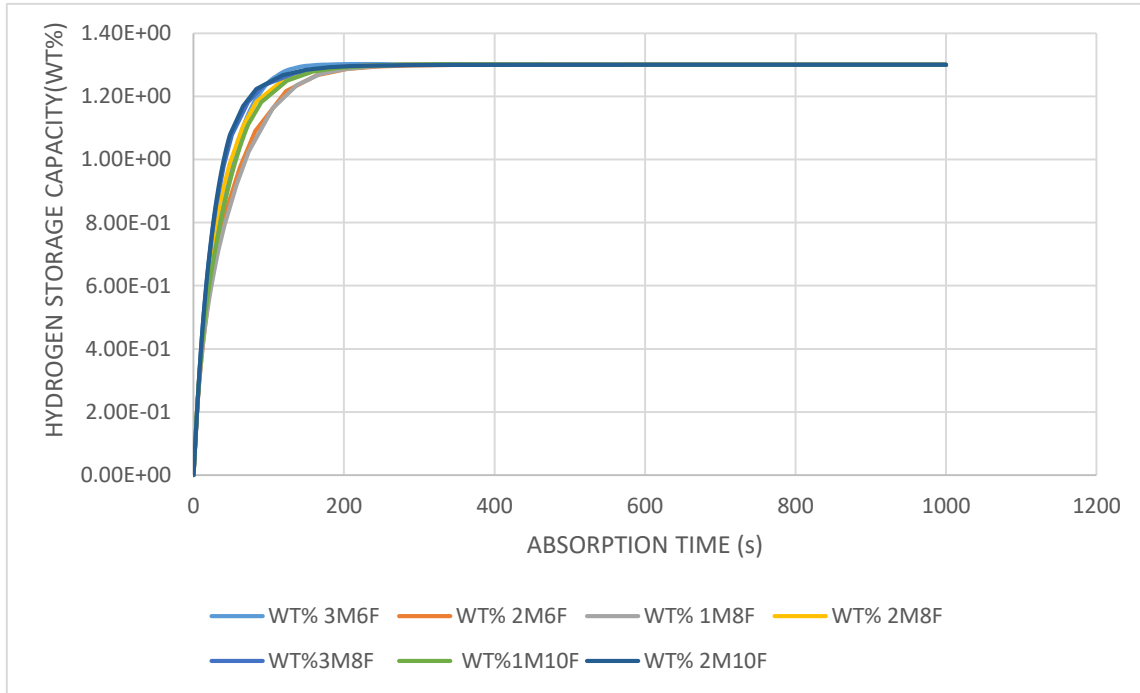
Fig.2 Effect of number of fins each with thickness 2mm on variations of (a) Average bed temperature (b) Hydrogen storage capacity during hydriding process with time

4.1.b EFFECT OF THICKNESS OF FINS

By keeping the number of fins constant, when the thickness is increased, it was observed that the absorption time decreases as the maximum average temperature reached decreases. This happens because as the volume of the fin is increasing the cross-sectional area for heat propagation increases as a result more heat can be carried away to the HTF. At the same time with an increase in thickness the weight increase significantly. So to avoid a larger increase in weights as well as longer time for absorption, the following combinations are taken 6 fins of 2mm and 3mm, 8 fins of 1mm, 2mm and 3mm, 10 fins of 1mm and 2mm. The analysis results were plotted on the graph and the average bed temperature vs absorbing time is shown in **Fig.3(a)** and the Hydrogen capacity vs absorbing time graph for the above-mentioned combinations is shown in **Fig.3(b)**. From the graphs, it is clear that 8 fins of 2mm or 10 fins of 1mm have almost the same absorption time. Considering the weight factor 10 fin with 1mm thickness is selected for further analysis with varying convective heat transfer coefficient.



(a)



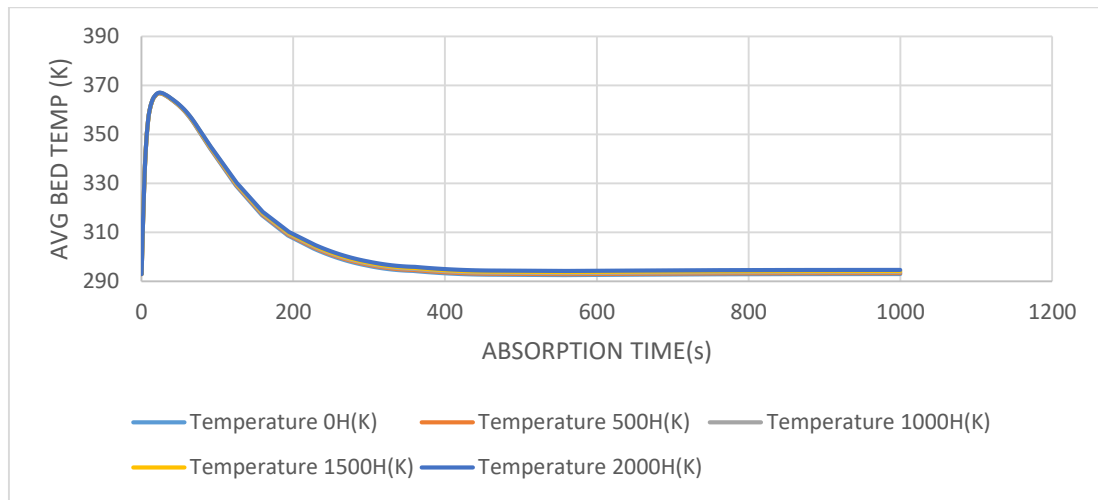
(b)

Fig.3 Effect of fin thickness on variations of (a) Average bed temperature (b) Hydrogen storage capacity during hydriding process with time

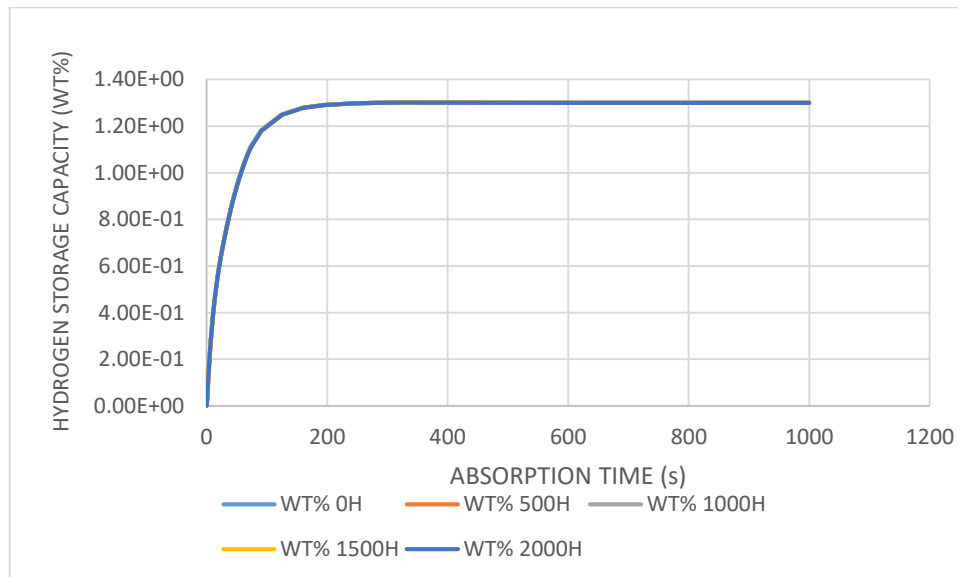
4.1.c EFFECT OF CONVECTIVE HEAT TRANSFER COEFFICIENT OF SHELL

By increasing the convective heat transfer coefficient of the shell from 0-2000W/m²K (0,500,1000,1500,2000) for 10 fins of 1mm thickness MH based Hydrogen storage device, it

is observed that there is a very little increase in heat absorption capacity. The Average bed temperature and Hydrogen absorption capacity with time by changing convective heat transfer coefficient is shown in **Fig.4 (a)** and **(b)** respectively. As the convective Heat transfer coefficient increase more amount of heat can be transferred to the atmosphere, but a very slight variation is observed because the partial pressure of Hydrogen in the MH bed region near the shell is low as a result absorption is also less and hence heat generated is low and convective heat transfer coefficient has no great impact. Therefore, the outer surface of shell doesn't require any forced convective cooling methods.



(a)



(b)

Fig.4 Effect of convective heat transfer coefficient of the outer surface of shell on variations of (a) Average bed temperature (b) Hydrogen storage capacity during hydriding process with time

4.1.d. PICTORIAL REPRESENTATION OF THE HYDRIDE BED TEMPERATURE DURING HYDRIDING PROCESS

Hydriding process at different time intervals of 0, 50, 100, 200, 300, and 500 s at an absorption condition of 30bar in the form of sliced view is presented in **Fig. 5**. The distance between the two slices is taken as 30 mm.

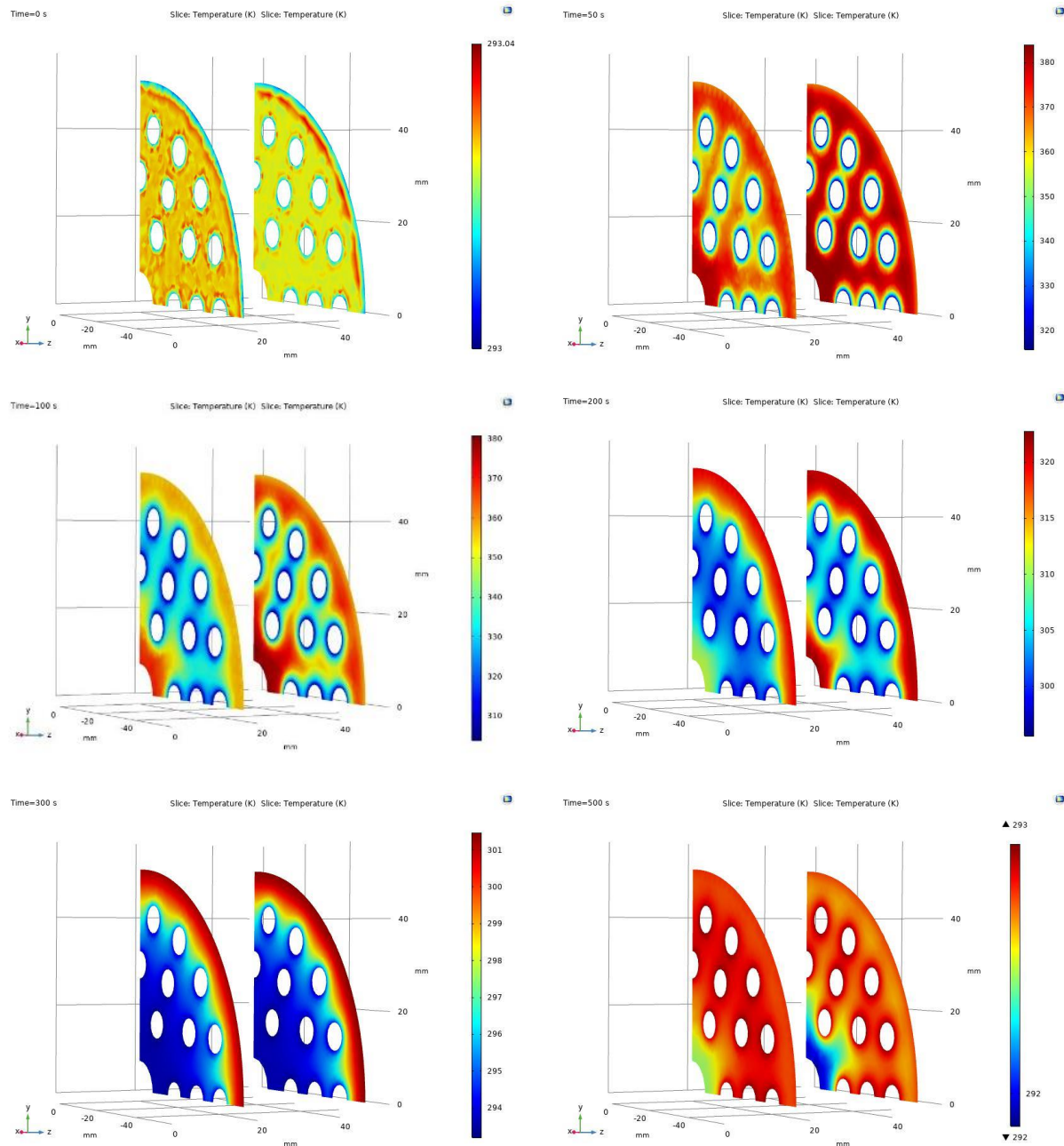


Fig. 5 Pictorial sliced views of hydride bed temperature at different time intervals during hydriding process.

4.2. CONCLUSIONS

The transient behavior of heat and hydrogen transfer characteristics of $\text{LaNi}_{4.91}\text{Sn}_{0.15}$ based hydrogen storage container is studied using COMSOL Multiphysics 5.6. A 2-D mathematical model is developed for a cylindrical reactor representing the cross-section of the hydride bed with 60 ECT for understanding the effect of absorption of Hydrogen. This study is then extended to a 3-D mathematical model employing 2.25kg of $\text{LaNi}_{4.91}\text{Sn}_{0.15}$ based storage container with 36 ECT and their hydriding characteristics are predicted for different number of fins (4-10), different thickness of the fins (1-3mm), and different convective heat transfer coefficient of the shell. The variations in hydrogen storage capacity (wt%) are reported for the above-mentioned conditions. For the reactor with 8 fins each of 2mm thickness and 10 fins of 1mm thickness, the rate of hydriding is found to be rapid for the supply condition of 30 bar hydrogen supply pressure considering the weight factor, 10 fins each of 1mm is selected as the best option. For the hydriding process, with the increase in the convective heat transfer coefficient of the shell, the absorption rate increases but very slowly. Therefore, natural convection is preferred over forced convection to save energy.

4.3. FUTURE SCOPE

Further development in the design of MH-based Hydrogen storage device can store the eco-friendly hydrogen efficiently. And MH reactors can be manufactured for large-scale applications like heat pumps, refrigerators, hydrogen fuel, heat transformers, etc.

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